

Exercise 6.1 (Deterministic channel) Consider a memoryless channel that takes two bits (X_1, X_2) as input and maps them to two bits of output (Y_1, Y_2) as follows: $00 \rightarrow 01$, $01 \rightarrow 10$, $10 \rightarrow 11$, $11 \rightarrow 00$.

- Calculate the mutual information $I(X_1, X_2; Y_1, Y_2)$ for a given joint pmf of the four pairs of input bits. You can express your answer in terms of $p_{ij} = P(X_1 = i, X_2 = j)$ for $i, j \in \{0, 1\}$.
- Show that the channel mutual information is 2.
- Show that, surprisingly, $I(X_1; Y_1) = 0$ for the distribution that achieves the maximal mutual information in part (b) (that is, information is only transferred by considering both bits).
Hint: Find the joint pmf of X_1 and Y_1 .

Solution:

- Since the channel is deterministic, we have

$$I(X_1, X_2; Y_1, Y_2) = H(Y_1, Y_2) - H(Y_1, Y_2 | X_1, X_2) = H(Y_1, Y_2) - 0 = H(Y_1, Y_2).$$

Given input distribution $(p_{00}, p_{01}, p_{10}, p_{11})$, the output distribution can be expressed as $(p_{11}, p_{00}, p_{01}, p_{10})$. Thus,

$$I(X_1, X_2; Y_1, Y_2) = H(p_{11}, p_{00}, p_{01}, p_{10}) = - \sum_{i,j \in \{0,1\}} p_{ij} \log p_{ij}.$$

- The maximizing input distribution is clearly uniform, *i.e.*,

$$p_{ij}^* = 1/4 \quad \forall (i, j) \in \{0, 1\}^2.$$

Thus,

$$C = H(p_{11}^*, p_{00}^*, p_{01}^*, p_{10}^*) = 4 \cdot \left(-\frac{1}{4} \log_2 \frac{1}{4} \right) = 2 \text{ bits per channel use.}$$

- Note that

$$Y_1 = X_1 \oplus X_2.$$

We now calculate the conditional probability $p_{Y_1|X_1}$ directly. Consider

$$\begin{aligned} p_{Y_1|X_1}(y_1|0) &= \sum_{x_2} p_{Y_1|X_1 X_2}(y_1|0, x_2) p_{X_2|X_1}(x_2|0) \\ &= \sum_{x_2} p_{Y_1|X_1 X_2}(y_1|0, x_2) p_{X_2}(x_2). \end{aligned}$$

For $y_1 = 0$, we have

$$\begin{aligned} p_{Y_1|X_1}(0|0) &= \frac{1}{2} p_{Y_1|X_1 X_2}(0|0, 0) + \frac{1}{2} p_{Y_1|X_1 X_2}(0|0, 1) \\ &= \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 0. \end{aligned}$$

Hence,

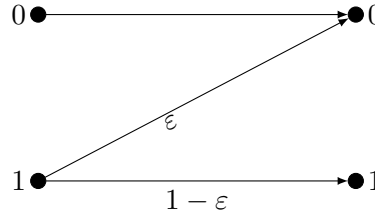
$$p_{Y_1|X_1}(y_1|0) = \begin{cases} 1/2 & y_1 = 0, \\ 1/2 & y_1 = 1. \end{cases}$$

Similarly,

$$p_{Y_1|X_1}(y_1|1) = \begin{cases} 1/2 & y_1 = 0, \\ 1/2 & y_1 = 1. \end{cases}$$

Therefore, using the capacity achieving distribution, the sub-channel is completely noisy, and $I(X_1, Y_1) = 0$.

Exercise 6.2 The Z channel is a binary channel with conditional pmf $p(0|0) = 1$, $p(0|1) = \epsilon$ and $p(1|1) = 1 - \epsilon$.



a) Suppose $\epsilon = 1/2$, compute the channel mutual information.

b) Plot the channel mutual information as a function of $\epsilon \in [0, 1]$.

Hint: You will need to perform some numerical optimisations here.

Solution:

a.) Let $P_X(0) = \alpha$ and $P_X(1) = 1 - \alpha$. The channel mutual information can be calculated as follows:

$$\begin{aligned} I(W) &:= \max_{\alpha} H(Y) - H(Y|X) \\ &= \max_{\alpha} H(Y) - \sum_{x=0,1} H(Y|X=x)P_X(x) \\ &= \max_{\alpha} H_b(\epsilon + (1 - \epsilon)\alpha) - H(Y|X=1)P_X(1) \\ &= \max_{\alpha} H_b(\epsilon + (1 - \epsilon)\alpha) - (1 - \alpha)H_b(\epsilon) \\ &= \max_{\alpha} -(\epsilon + (1 - \epsilon)\alpha) \log(\epsilon + (1 - \epsilon)\alpha) - (1 - \epsilon)(1 - \alpha) \log(1 - \alpha) + (1 - \alpha)\epsilon \log \epsilon. \end{aligned}$$

To find the maximum, we take the derivative of the above expression with respect to α and set it to zero.

$$\frac{dI(W)}{d\alpha} = -(1 - \epsilon) \log(\epsilon + (1 - \epsilon)\alpha) + (1 - \epsilon) \log(1 - \alpha) - \epsilon \log \epsilon.$$

By calculus, the derivative is 0 when

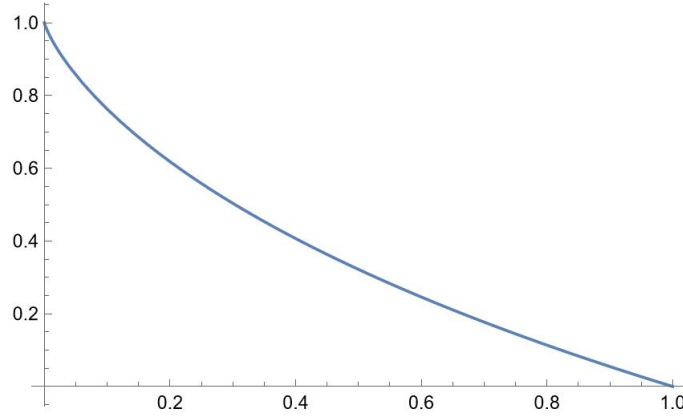
$$\alpha^* = \frac{\epsilon^{-\frac{\epsilon}{1-\epsilon}} - \epsilon}{\epsilon^{-\frac{\epsilon}{1-\epsilon}} - \epsilon + 1}$$

and is positive for $\alpha < \alpha^*$ and negative for $\alpha > \alpha^*$. By setting $\epsilon = \frac{1}{2}$,

$$\alpha^* = 1 - \frac{1}{(1/2)(1 + 2^2)} = \frac{3}{5},$$

and thus

$$I(W) = H_b(1/5) - 2/5.$$



b.) The plot is given here.

Exercise 6.3 For two positive integers k and m , let $k \bmod m$ be the remainder when k is divided by m . Find the channel mutual information of the m -input discrete memoryless channel in which

$$Y = X + Z \bmod m,$$

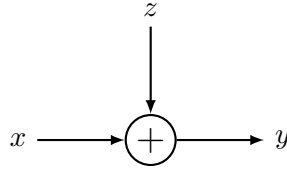
where $X \in \{0, 1, \dots, m-1\}$, $P[Z = 1] = \frac{3}{4}$, and $P[Z = 0] = \frac{1}{4}$.

Solution: The capacity is $\log_2 m - h(1/4)$. Consider

$$I(X; Y) = H(Y) - H(Y|X) = H(Y) - h(1/4).$$

Note that $H(Y)$ is maximized at the value $\log_2 m$ and this is achievable using the uniform input distribution $p(x) = 1/m$ for all $x \in \{0, 1, \dots, m-1\}$.

Exercise 6.4 Find the channel mutual information of the following channel:



Here, the random variable Z is independent of X and $P[Z = 0] = P[Z = a] = \frac{1}{2}$ for some real number a . Further assume that X is binary.

Solution: We can identify two different values for the capacity of the channel according to the value of a .

- a.) When $a \neq \pm 1$, the outputs of the channel are non-overlapping. In such cases the channel is a noisy channel with non-overlapping outputs. It is known that the capacity of such a channel is 1 bit since:

$$C = \max_{p_X} I(X; Y) = \max_{p_X} H(X) - H(X|Y) = \max_{p_X} H(X) = 1$$

where the third equality follows from the non-overlapping fact (conditioning $Y = y$, $X|Y = y$ is deterministic).

- b.) When $a = 1$ or -1 , the outputs of the channel are overlapping. If $a = 1$, Y can be 1 for either inputs, but is 0 (or 2) only if $X = 0$ (or 1). If $a = -1$, Y can be 0 for either inputs,

but is -1 (or 1) only if $X = 0$ (or 1). We compute the channel capacity for $a = 1$. By symmetry, the capacity is the same when $a = -1$.

$$\begin{aligned} H(Y) &= -p_Y(0) \log_2 p_Y(0) - p_Y(1) \log_2 p_Y(1) - p_Y(2) \log_2 p_Y(2) \\ &= -\frac{p_X(0)}{2} \log_2 \frac{p_X(0)}{2} - \left(\frac{p_X(0)}{2} + \frac{p_X(1)}{2} \right) \log_2 \left(\frac{p_X(0)}{2} + \frac{p_X(1)}{2} \right) - \frac{p_X(1)}{2} \log_2 \frac{p_X(1)}{2} \\ &= \frac{1}{2} H(X) + 1, \end{aligned}$$

and

$$H(Y|X) = p_X(0)H(Y|X=0) + p_X(1)H(Y|X=1) = p_X(0) + p_X(1) = 1.$$

Thus,

$$C = \max_{p_X} I(X; Y) = \max_{p_X} H(Y) - H(Y|X) = \frac{1}{2} \max_{p_X} H(X) = \frac{1}{2} \text{ bits per channel use.}$$

Exercise 6.5 Let X and Z be independent random variables taking values on $\{1, \dots, n\}$ and $\{0, 1\}$, respectively, with $p_X(i) = q_i$ (for each i) and $p_Z(1) = p$. Define the random variable $Y := X \cdot Z$.

- Write $H(Y)$ in terms of $H(X)$ and $H(Z)$.
- Find p and $\mathbf{q} = (q_1, \dots, q_n)$ that maximize $H(Y)$.
- Suppose X and Y are input and output of a channel. For a fixed $p \in [0, 1]$, what is the channel mutual information $I(p)$?

Solution:

- Note that the pmf of Y can be written as

$$P_Y(y) = \begin{cases} 1-p, & y=0 \\ pq_i, & y=1, 2, \dots, n \\ 0, & \text{otherwise.} \end{cases}$$

Thus,

$$\begin{aligned} H(Y) &= (1-p) \log \frac{1}{1-p} + \sum_{i=1}^n pq_i \log \frac{1}{pq_i} \\ &= (1-p) \log \frac{1}{1-p} - p \log p \sum_{i=1}^n q_i - p \sum_{i=1}^n q_i \log q_i \\ &= H(Z) + pH(X). \end{aligned}$$

- Without loss of generality, we measure in nats in this question. Note that $H(Y) = h(p) + p \cdot H(X)$, where h is the binary entropic function (in nats). One must have $H(Y) \leq h(p) + p \log n$ since $H(X) \leq \log n$, and the equality is attained when $\mathbf{q} = (\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n})$. Taking derivatives of $h(p) + p \log n$ with respect to p , we have

$$\frac{d}{dp} H(Y) = \log \frac{1-p}{p} + \log n,$$

which is 0 when $p = p^* = \frac{n}{n+1}$, positive when $p < p^*$, and negative when $p > p^*$. As a result,

$$h(p) + p \log n \leq h\left(\frac{n}{n+1}\right) + \frac{n}{n+1} \log n.$$

Thus,

$$H(Y) \leq h\left(\frac{n}{n+1}\right) + \frac{n}{n+1} \log n$$

with equality when $\mathbf{q} = (\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n})$ and $p = \frac{n}{n+1}$.

c.)

$$\begin{aligned} I(p) &:= \max_{\mathbf{q}} I(X : Y) \\ &= \max_{\mathbf{q}} H(Y) - H(Y|X) \\ &= \max_{\mathbf{q}} H(Z) + pH(X) - H(XZ|X) \\ &= \max_{\mathbf{q}} H(Z) + pH(X) - H(Z) \\ &= \max_{\mathbf{q}} pH(X) \\ &= p \log n. \end{aligned}$$

Exercise 6.6 (Composed channels) Consider two channels $W_1(y_1|x_1)$ and $W_2(y_2|x_2)$ with channel mutual information I_1 and I_2 , respectively. Find the channel mutual information of the following composed channels.

a) *Parallel composition:*

$$W(y_1, y_2|x_1, x_2) = W_1(y_1|x_1)W_2(y_2|x_2).$$

b) *Selecting channel based on another input bit z :*

$$W(y|x, z) = \begin{cases} W_1(y|x) & \text{if } z = 0 \\ W_2(y|x) & \text{if } z = 1 \end{cases}.$$

Here we assume that either $W_1(y|x) = 0$ or $W_2(y|x) = 0$ for all x and y , that is, the channel outputs of the two channels are orthogonal.

Solution:

a.) Firstly, note that the new channel has input alphabet $\mathcal{X}_1 \times \mathcal{X}_2$, output alphabet $\mathcal{Y}_1 \times \mathcal{Y}_2$ and channel law

$$P_{Y_1 Y_2 | X_1 X_2}(y_1, y_2|x_1, x_2) = P_{Y_1 | X_1}(y_1|x_1)P_{Y_2 | X_2}(y_2|x_2)$$

This is a *product channel*. So we must find a input distribution on $\mathcal{X}_1 \times \mathcal{X}_2$ that maximizes $I(X_1 X_2; Y_1 Y_2)$. Notice that since the joint distribution of (X_1, X_2, Y_1, Y_2) factorizes as

$$P_{X_1 X_2}(x_1, x_2)P_{Y_1 Y_2 | X_1 X_2}(y_1, y_2|x_1, x_2) = P_{X_1 X_2}(x_1, x_2)P_{Y_1 | X_1}(y_1|x_1)P_{Y_2 | X_2}(y_2|x_2)$$

we have $Y_1 - X_1 - X_2 - Y_2$ forming a Markov chain.

Let us first find an upper bound on the mutual information. We have

$$\begin{aligned} I(X_1 X_2 : Y_1 Y_2) &= H(Y_1 Y_2) - H(Y_1 Y_2 | X_1 X_2) \\ &= H(Y_1 Y_2) - H(Y_1 | X_1 X_2) - H(Y_2 | X_1 X_2) \\ &= H(Y_1 Y_2) - H(Y_1 | X_1) - H(Y_2 | X_2) \\ &\leq H(Y_1) + H(Y_2) - H(Y_1 | X_1) - H(Y_2 | X_2) \\ &= I(X_1 : Y_1) + I(X_2 : Y_2), \end{aligned}$$

where the second and third equalities follow from the Markov chain $Y_1 - X_1 - X_2 - Y_2$. Note that equality in the above chain of inequalities holds when X_1 and X_2 are independent. Hence,

$$\begin{aligned} C &:= \max_{P_{X_1 X_2}} I(X_1 X_2 : Y_1 Y_2) \\ &\leq \max_{P_{X_1 X_2}} I(X_1 : Y_1) + I(X_2 : Y_2) \\ &= \max_{P_{X_1}} I(X_1 : Y_1) + \max_{P_{X_2}} I(X_2 : Y_2) \\ &= C_1 + C_2, \end{aligned}$$

with equality iff $P_{X_1 X_2} = P_{X_1}^* P_{X_2}^*$ where $P_{X_j}^*$ maximizes the mutual information of the j -th channel.

Thus the input distribution that maximizes $I(X_1 X_2 : Y_1 Y_2)$ is of the product form $P_{X_1 X_2}^* = P_{X_1}^* P_{X_2}^*$ where the constituent distributions $\{P_{X_j}^*\}_{j=1,2}$ are those maximizing the mutual information of the j -th channel. The channel mutual information, *i.e.*, the maximal value of $I(X_1 X_2 : Y_1 Y_2)$, is $C_1 + C_2$.

b.) Suppose that

$$p = \sum_y P(Y = y, Z = 0).$$

Using the chain rule, we have

$$I(Y : XZ) = I(Y : Z) + I(Y : X|Z).$$

On the one hand, as the channel output when $Z = 0$ and $Z = 1$ are orthogonal, we can infer Z from Y with certainty,

$$I(Y : Z) = H(Z) - H(Z|Y) = H(Z) = h(p).$$

On the other hand, we can easily obtain

$$I(Y : X|Z) = pI(Y : X|Z = 0) + (1 - p)I(Y : X|Z = 1).$$

Therefore

$$I(Y : XZ) = h(p) + pI(Y : X|Z = 0) + (1 - p)I(Y : X|Z = 1).$$

By maximizing both sides,

$$\begin{aligned} I(W) &= \max_{P(XZ)} I(Y : XZ) = \max_{P(XZ)} (h(p) + pI(Y : X|Z = 0) + (1 - p)I(Y : X|Z = 1)) \\ &\leq \max_p (h(p) + p \max_{P(X|Z=0)} I(Y : X|Z = 0) + (1 - p) \max_{P(X|Z=1)} I(Y : X|Z = 1)) \\ &\leq \max_p (h(p) + pI_1 + (1 - p)I_2). \end{aligned}$$

Setting the derivative, we obtain

$$\frac{d}{dp} (h(p) + pI_1 + (1 - p)I_2) = -\log_2 \frac{p}{1 - p} + (I_1 - I_2) = 0 \implies p^* = \frac{1}{1 + 2^{I_2 - I_1}},$$

and thus we can upper bound $I(W)$ with

$$I(W) \leq I_1 + \log_2(1 + 2^{I_2 - I_1}).$$

Furthermore, we will show that the following probability distribution achieves the upper bound:

$$P^*(X = x, Z = 0) = p^* P_1^*(x), \quad P(X = x, Z = 1) = (1 - p^*) P_2^*,$$

where P_1^* and P_2^* are the probability distributions which achieve the channel mutual information I_1 and I_2 of W_1 and W_2 . In this case,

$$I(Y : XZ) = I(Y : Z) + I(Y : X|Z) = h(p^*) + p^* I_1 + (1 - p^*) I_2 = I_1 + \log_2(1 + 2^{I_2 - I_1}).$$

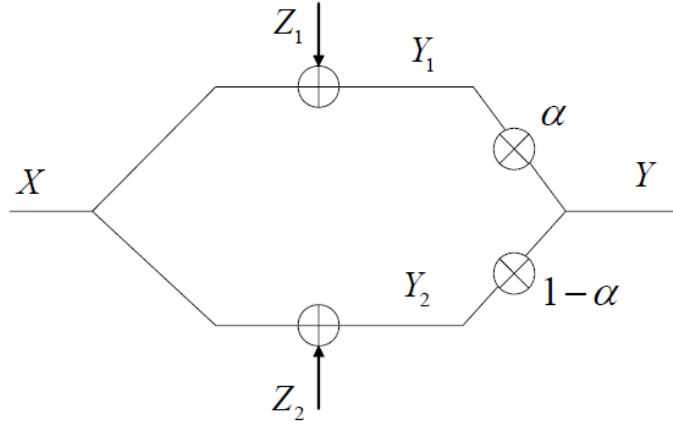
Therefore, the channel mutual information of the composite channel is

$$I(W) = I_1 + \log_2(1 + 2^{I_2 - I_1}).$$

Exercise 6.7 Consider a Gaussian channel shown below, in which the transmitted signal X with $\mathbb{E}[X^2] \leq P$ is received by two antennas with $Y_1 = X + Z_1$ and $Y_2 = X + Z_2$ where Z_1 and Z_2 are independent with $\mathbb{E}[Z_i^2] = \sigma_i^2$ ($\sigma_1^2 < \sigma_2^2$). The signals at the two antennas are combined as

$$Y = \alpha Y_1 + (1 - \alpha) Y_2$$

before decoding, where $0 \leq \alpha \leq 1$.



Find the power-constrained channel mutual information of this channel as a function of $\alpha \in [0, 1]$.

Solution: This is a Gaussian channel with signal power P and noise power

$$\alpha^2 \sigma_1^2 + (1 - \alpha)^2 \sigma_2^2.$$

Hence, the capacity is

$$\frac{1}{2} \log \left(1 + \frac{P}{\alpha^2 \sigma_1^2 + (1 - \alpha)^2 \sigma_2^2} \right) \quad \text{bits per channel use.}$$

The maximum capacity is reached for $\alpha^* = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$, and is given by $C^* = \frac{1}{2} \log \left(1 + \frac{P}{\sigma_1^2} + \frac{P}{\sigma_2^2} \right)$ bits per channel use.